

Stiefel and Grassmann Manifolds

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Abstract

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1 Introduction

2 Preliminaries

We follow the conventions of [2] and [1].

2.1 Basic Definitions

Definition 2.1.

Definition 2.2.

2.2 Examples

Example 2.3.

Example 2.4.

3 Motivation

4 The Stiefel Manifold

Definition 4.1.

Remark 4.2.

5 The Grassmann Manifold

Lemma 5.1.

Proof. □

Proposition 5.2.

Proof. □

6 From Stiefel to Grassmann

Theorem 6.1.

Proof. □

Corollary 6.2.

Proof. □

7 Worked Example: $\text{Gr}(2, 4)$

7.1 First Application

7.2 Second Application

8 Discussion

9 Further Directions

10 Conclusion

References

- [1] Nicolas Boumal, *An introduction to optimization on smooth manifolds*, Cambridge University Press, 2023.
- [2] Ed Segal, *Manifolds*, 2016, Lecture notes, University College London.